

SDMC: Offering premium home entertainment experiences with easy-to- deploy OTT solutions

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About SDMC

To offer an easy-to-deploy, high-performance solution, SDMC leverages the computing, video processing and data analytics capabilities of Google Cloud to reduce deployment time

SDMC is a global leader in delivering OTT devices, broadband devices and SaaS services to operators worldwide. In 2017, SDMC launched XMediaTV, a one-stop OTT solution that integrates various video content sources to provide smooth TV watching experiences across all devices. SDMC's OTT solution is currently adopted by more than 150 operators around the world.

Industries: Media & Entertainment

enhance TV watching experiences.

Location: China

Google Cloud results

- Shortens live video streaming latency by more than 50% with Media CDN
- Reduces operations workload while supporting instant scalability through GKE
- Supports near real-time data analytics for video recommendations with BigQuery
- Enables personalized ad insertion in live programs for higher ad revenue

Should solve deployment time from six months to days

(<https://cloud.google.com/media-cdn>)

Google Cloud (<https://cloud.google.com/>)

BigQuery (<https://cloud.google.com/bigquery/>)

Kubernetes Engine

(<https://cloud.google.com/kubernetes-engine/>)

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Media CDN (<https://cloud.google.com/media-cdn>)

As OTT video content becomes increasingly popular, telecom and TV operators are fiercely competing

against each other to offer the best over-the-top (OTT) service, a service that distributes video content via the internet. To successfully build a successful OTT service, operators normally need to work with different providers to acquire resources for video transcoding, video encryption, device management and content delivery network.

SDMC (<https://en.sdmctech.com/about/about-us.html>) is on a mission to build a smart home ecosystem by helping global operators deliver smart home media entertainment services in the most effortless way. Since 2003, the video technology company has been developing and offering various innovative hardware and software products for digital TV and smart homes. In 2017, SDMC launched XMediaTV, a one-stop OTT solution for Android TV devices that integrates leading OTT platforms with Google services ranging from Android TV, Widevine, Google Play (<https://play.google.com/store/games>), YouTube (<https://www.youtube.com/>) to Google Assistant (<https://assistant.google.com/>). As of August 2022, SDMC's OTT solution has been adopted by more than 150 operators in Southeast Asia, Middle East, Europe, North and South America.

"We want to propose an OTT solution that customers can deploy in the shortest time possible to provide quality services," Kevin Zhang, Vice President & Product Director at SDMC. "Also, since we have integrated the Google ecosystem into our solution and offer hardware products like Android TVs, operators don't need to look further to build an all-inclusive and advanced smart TV service for their customers at lower costs."

In the past, SDMC deployed XMediaTV in its customers' on-prem environments. The deployment process was laborious and time consuming, because operators had to purchase hardware by themselves, and SDMC needed to send its engineers to complete the setup. The costs of project launch, implementation and maintenance were very high. Furthermore, using on-prem IT infrastructure translated into limited scalability. Customers sometimes were not able to meet the increasing demand or handle flow surges, which resulted in service disruptions and undermined the TV watching experience that they offered. The SDMC team then realized that to reduce operational costs for its customers and improve service quality, it needed to run XMediaTV in public cloud, so that operators could launch the service at any time, use resources according to their needs, pay for resources according to their use, avoid maintenance workloads and focus more on their business. To that end, SDMC landed on Google Cloud (<https://cloud.google.com/>) in early 2022, because it supports the Google ecosystem that XMediaTV was built upon, and can be seamlessly integrated.

"We did an in-depth study of the products of Google Cloud and found that Google Cloud not only has highly scalable computing resources and powerful video processing tools, but also fits our solution that is based on the Google ecosystem the best," explains Zhang. "At the same time, we've fully integrated our product with Android TV to offer a one-stop OTT solution for operators. Google Cloud is the most ideal choice for us to improve our OTT product in all respects."

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**Facilitating OTT solution
deployment with powerful video
processing tools**

SDMC now uses Live Stream API

(<https://cloud.google.com/livestream>) to process live streaming videos for output. The company also employs Transcoder API

(<https://cloud.google.com/transcoder/docs>) to transcode on-demand videos and add watermarks to live streaming videos. Zhang says that using cloud video processing tools has enabled SDMC to set up media streaming services for its customers in a short period of time, because it doesn't involve any dedicated hardware, and Transcoder API can transcode videos three times faster than the transcoding solution that the company previously used. Before, it took three to six months for SDMC to deploy XMediaTV for its customers. Now, it only needs a few days to complete the whole deployment process.

On top of that, SDMC's operator customers can now expand their services without worrying about insufficient computing resources. When XMediaTV was deployed in on-prem environments, SDMC used an open-source tool to add watermarks to live streaming programs. The watermarking process required a lot of CPU resources. SDMC's customers therefore needed to balance between the computing resource cost and the number of live programs that they could offer. With Transcoder API, operators can process as many live streaming videos as they want, which is particularly beneficial for smaller operators with limited budgets.

"Live Stream API and Transcoder API have helped bring our OTT solution to a new level by significantly shortening deployment time and providing unlimited

resources for video processing," says Zhang. "We're now able to help operators of all sizes build quality OTT services."

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Enhancing TV watching experiences while reducing operational costs

By adopting the infrastructure of Google Cloud, SDMC has successfully enhanced the quality of its OTT services and reduced operational costs. To deliver video content around the world, the company uses Media CDN (<https://cloud.google.com/media-cdn>), which provides 10X more edge nodes that support faster and more stable global data transfer. Before the adoption of Google Cloud, there used to be a three to five-second latency for video streams on XMediaTV. Thanks to the extensive network offered by Media CDN, SDMC is able to reduce the streaming latency to just one to two seconds.

The SDMC team has been leveraging Kubernetes clusters to run XMediaTV and now uses Google Kubernetes Engine (<https://cloud.google.com/kubernetes-engine>) (GKE) to deploy the clusters. The autoscaling feature of GKE means that SDMC's customers can always meet the viewing demand instantly. In on-prem environments, operators had to add hardware to scale up clusters and were sometimes forced to limit the audience number for popular live broadcast shows, because there were no additional computing resources available. In addition to not having to limit the number of audiences, SDMC's operations workload has been greatly reduced because of the fully-

managed GKE clusters. Zhang says that before, SDMC needed to dispatch one engineer to each customer for daily system maintenance. Now, only three engineers are required to cover the operations work for all its customers.



Increasing data analytics efficiency with BigQuery

Data analytics is an important part of SDMC's services, as its customers rely on business intelligence reports to improve their operations. Previously, SDMC developed its data analytics platform with four to five engineers overseeing the process. Even then, the analytics efficiency varied greatly depending on its customers' hardware capacity. By leveraging BigQuery (<https://cloud.google.com/bigquery>) to analyze data in near real-time, SDMC is able to guarantee high data analytics performance for all its customers and shorten the time needed to generate analytics results.

"With the excellent data analytics capabilities supported by BigQuery, we are able to effortlessly

provide our customers with any data insights they need for their business. We no longer need engineers to develop our own data analytics functionalities and now have more staff dedicated to our media services," notes Zhang.

Enabling personalized advertising to boost monetization

SDMC understands that advertising is crucial to its customers' business, which is why it has also improved the advertising capabilities of XMediaTV when adopting Google Cloud. Before, the company relied on the client-side ad insertion (CSAI) technology to display ads from [Ad Manager](https://admanager.google.com/home/) (<https://admanager.google.com/home/>) and [Ad Mob](https://admob.google.com/home) (<https://admob.google.com/home>) on XMediaTV. Users could easily block ads by using ad blockers, and it was not possible to show personalized ads. Through [Video Stitcher API](https://cloud.google.com/video-stitcher/docs) (<https://cloud.google.com/video-stitcher/docs>), SDMC is able to leverage the server-side ad insertion (SSAI) technology to prevent ad blocking, and insert personalized ads into on-demand videos and live streams, which results in higher ad conversion rates and revenue.

"As advertising is the lifeblood of TV operators' revenue, many of our customers need an effective and easy-to-use advertising system for their OTT services," says Zhang. "Video Stitcher API has helped us overcome the technical limitations to enhance ad performance for our customers. This way, they can easily increase ad revenue without

coming up with new monetization strategies themselves."

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Offering a comprehensive smart home solution with AI

Next, SDMC is looking to leverage the artificial intelligence (AI) capabilities of Google Cloud to offer more comprehensive smart TV and smart home services. The company will use Translation AI (<https://cloud.google.com/translate>) to swiftly offer content and subtitle translations for operators that provide services in different languages, and employ Video AI (<https://cloud.google.com/video-intelligence>) to automatically identify illegal or inappropriate content uploaded to XMediaTV while classifying videos based on locations, actors or scenes for more precise video recommendations.

"We're impressed by how Google Cloud has upgraded our OTT solution and made it possible for our customers to offer premium TV watching experiences with minimum effort," says Zhang. "With the wide-ranging capabilities of Google Cloud, we're ready to bring our smart TV and smart home services to the next level."